

Microplastic Degradation Research Project

Phase 1 — Pepsin Cleavage Site Mapping

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This document records Phase 1: retrieving the wild-type IsPETase sequence from UniProt, applying the ExPASy pepsin pH 1.3 cleavage rule to identify all 35 sequence-level pepsin cleavage sites, and annotating each site with local sequence context and amino acid category.

The Biology — What Phase 1 Answers

Why pepsin matters

Pepsin is the primary protease secreted by the stomach lining at pH 1.2-2.0. It acts as molecular scissors that cuts proteins at specific sequences — preferentially at Phenylalanine (F) and Leucine (L) residues. Wild-type IsPETase has not evolved any resistance to gastric proteolysis, making pepsin cleavage one of the two primary failure mechanisms under gastric conditions.

The ExPASy pepsin pH 1.3 rule

The ExPASy pepsin rule at pH 1.3 identifies all F and L residues where cleavage is predicted based on the surrounding sequence context. This rule was implemented via the pyteomics library and applied to the full 290 amino acid IsPETase sequence to identify every predicted cleavage site.

Phase 1 Results

Sequence retrieved

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UniProt accession : A0A0K8P6T7
Protein name      : IsPETase (PET hydrolase, Ideonella sakaiensis 201-F6)
Sequence length  : 290 amino acids
Download method   : urllib.request from rest.uniprot.org
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Pepsin cleavage sites found: 35 total

Category	Count	Significance
Leucine (L) P1 sites	14	Most common high-risk site type
Phenylalanine (F) P1 sites	5	Aromatic residue, high pepsin affinity
Other P1 sites	16	Lower risk, context-dependent
TOTAL	35	All annotated in output TSV file

Key finding: Of the 35 predicted pepsin cleavage sites, only L117 is surface-exposed (confirmed in Phase 2 with $\text{rel_ASA} = 0.348$) and a high-risk P1 residue. This makes L117 the single most important pepsin vulnerability to address. The conservative L117I substitution eliminates this site without structural disruption.

Output file produced

File: results/step1_pepsin/pepsin_cleavage_sites.tsv
Contents: position, P1 residue, peptide, context for all 35 sites